Salesforce Integration Analysis and Planning Report

Week 3 Internship Project | Prepared by Hrudhai Prem Kumar Patwari

SIMULATED TRAINING DESIGN. Cedarline Office Products and its system landscape are fictional. All flows, mappings, thresholds, and examples are proposals, not observed implementation evidence. No live Salesforce org, ERP, marketing platform, API credential, customer record, or company integration was accessed, configured, tested, or deployed.

Executive summary

This report proposes a specific integration design for Salesforce CRM, an ERP represented by SAP S/4HANA, and a marketing automation platform. The fictional distributor needs order visibility for sales and service, a controlled path to create ERP sales orders from approved opportunities, and campaign context that respects customer preferences. A middleware/iPaaS layer is proposed to validate, map, route, retry, and monitor messages. Order creation is asynchronous, keyed by stable identifiers, and paired with a durable acknowledgement; ERP order and shipment updates are returned to CRM. Marketing engagement flows both ways only for approved audience records, with preference suppression applied before activation. The supplied task includes no actual interface inventory, org metadata, or technical configuration, so the real current state is unknown. This report gives a discovery checklist, architecture, flows, field mappings, risk assessment, implementation phases, acceptance tests, and operating controls. Proposed latency goals are planning targets only and require stakeholder approval and end-to-end measurement. The immediate next step is to confirm data ownership, privacy rules, volumes, and service needs with sales, finance, marketing, security, and technical owners. Nothing here claims a deployed integration or hands-on Salesforce configuration.

1. Scope and current-state assessment

The assignment requires analysis of Salesforce with at least two external systems, documentation of requirements and data flows, diagrams including middleware and touchpoints, risk analysis, a practical implementation and maintenance plan, and a technical guide. This report uses an ERP and a marketing automation platform as the two candidate systems. No actual integration inventory, interface logs, contracts, record volumes, API consumption, or security policies were supplied. Consequently, current interfaces and current performance are unknown and are not represented as facts. The first project activity for a real organization must be discovery and validation.

Scenario assumptions

- Cedarline Office Products is a fictional B2B distributor for training purposes. Salesforce CRM is the proposed workspace for leads, accounts, contacts, opportunities, and cases.
- An ERP such as SAP S/4HANA is a candidate source for customer finance identifiers, sales orders, deliveries, and fulfillment status. ERP is tentatively authoritative for these operational facts.
- A marketing platform is a candidate for campaign membership, engagement, and preference data. The actual consent authority and precedence must be confirmed by the business and privacy owner.
- Middleware is a logical integration boundary. No vendor has been selected or procured.

Current-state discovery checklist

- Inventory inbound/outbound APIs, scheduled jobs, files, event subscriptions, business owners, credentials, and support contacts.
- Confirm objects/fields, external IDs, data classification, consent rules, record counts, peak loads, API capacity, and maintenance windows.
- Review failures, duplicates, retry behavior, timeouts, reconciliation evidence, incident history, and recovery procedures.
- Interview process owners; label each item as confirmed fact, assumption, or open question.

2. Proposed architecture

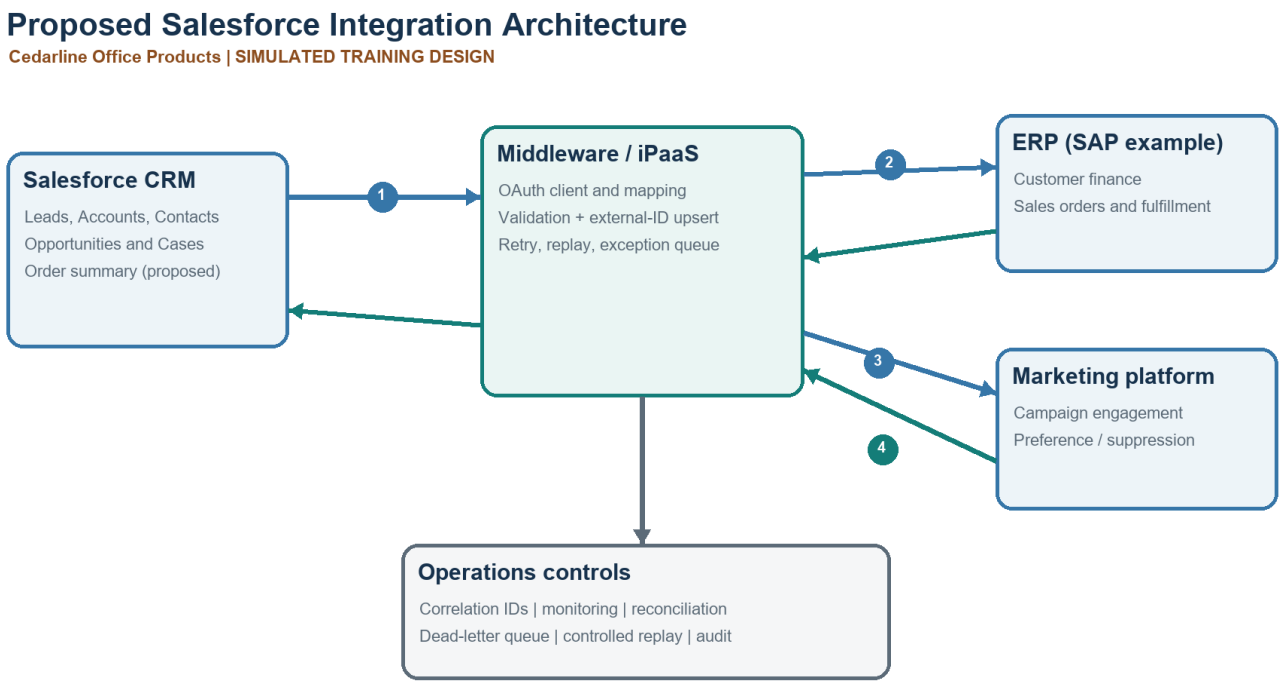


Figure 1. Target-state design, not a live-system export. Flow key: 1 Closed-Won order and eligible audience to middleware; 2 validated order to ERP and acknowledgement/status back; 3 consent-approved audience to marketing; 4 engagement and preference changes back to middleware and CRM.

Design rationale

Middleware separates user-facing CRM work from ERP and marketing systems and provides one place to validate payloads, transform values, manage credentials, route failures, and observe message state. Salesforce should not become the owner of ERP finance or fulfillment facts. Stable identifiers, timestamps, correlation IDs, schema versions, and idempotency keys support safe matching and replay. Reprocessing a message should not create a second order or contact.

3. Integration points and proposed flows

Flow	Trigger and direction	Candidate payload / match key	Pattern and proposed objective
A. Won opportunity to ERP order	Salesforce -> middleware -> ERP when a Closed Won opportunity receives required approvals.	Opportunity reference; ERP customer ID; approved ship-to; currency; line items; quantity/price; requested date; correlation ID.	Asynchronous request/reply with durable acknowledgement. Proposed: order acknowledgement within 5 min at p95, subject to approval and testing.
B. ERP order / shipment status	ERP -> middleware -> Salesforce when the order is accepted or delivery status changes.	ERP order ID; CRM opportunity reference; status; event timestamp; delivery reference; expected date. Match by immutable ERP order ID.	Event-driven or frequent incremental sync. Proposed: status visible within 15 min at p95.
C. Audience / campaign / preference	CRM -> middleware -> marketing for approved audience; marketing -> middleware -> CRM for engagement and preference updates.	CRM Lead/Contact ID; only approved contact fields; campaign ID/status; preference source, purpose, timestamp, and suppression flag.	Incremental async sync. Proposed: ordinary engagement <=30 min; suppression <=1 min if supported and approved.

Illustrative field ownership and mapping

Element	Authoritative source	Salesforce representation (proposed)	Control
ERP customer number	ERP	Account.ERP_Customer_ID__c (external ID)	ERP writes; immutable/unique; no fuzzy overwrite.
Opportunity/order reference	Salesforce opportunity + ERP order ID	Opportunity reference + ERP_Order_ID__c or Order_Summary__c	Carry both correlation keys; never create an order from name matching.
Order/delivery status	ERP	Order summary status + updated timestamp	ERP owns; upsert; reject older timestamps; expose freshness.
Campaign response	Marketing platform	Campaign Member status or approved equivalent	Map enumerated values; retain source/time; quarantine unknown values.
Marketing permission	Approved consent authority (confirm)	Preference status + source, purpose, timestamp	Apply approved precedence; suppress before outbound audience sync.

4. Technical requirements and design principles

Authorization. Use OAuth-based authorization, a dedicated integration identity, least-privilege object/field permissions, and protected secret storage. Salesforce documentation describes OAuth access through external client apps or connected apps and recommends external client apps for new app setup following Spring ’26 restrictions on creating new connected apps. Confirm current org policy before setup. [1]

API choice. Use standard REST operations for transactional records and external-ID upsert for matching. Use Bulk API 2.0 for large asynchronous loads or extracts rather than a row-by-row request loop. Salesforce describes Bulk API 2.0 as supporting asynchronous insert, upsert, query, and delete workloads. [2] Use an approved event approach for asynchronous notifications where event delivery semantics meet the business need; queued publication is not proof the downstream process completed. [3]

Reliability and data handling. Define timeouts, bounded retries with exponential backoff, retryable/permanent error classes, dead-letter handling, replay approvals, reconciliation, and alert owners. Validate schemas and required fields before transmission. Store correlation IDs and source timestamps; redact tokens and sensitive values from logs. Encrypt traffic, minimize personal data, rotate credentials, and set retention/deletion rules. API allocation and concurrency limits depend on org configuration and can change, so monitor current usage and test capacity before accepting throughput targets. [4]

5. Risk assessment

Risk	Likelihood / impact	Mitigation and evidence	Owner
Duplicate customer/order	Medium / High	External IDs, idempotency, replay test, reconcile order counts and values.	Integration + ERP
Unauthorized disclosure	Low-Med / Critical	Least privilege, OAuth, encrypted transport, secret vault, field minimization, access review, redacted logs.	Security
Latency / API throttling	Medium / High	Queueing, batching, backoff, capacity estimate, limit monitoring, freshness indicator.	Integration
Schema incompatibility	Medium / High	Versioned mappings, contract tests, notice period, backward-compatible changes.	System owners
Consent mismatch	Medium / Critical	Approved authority/precedence, suppression-first flow, source/time, negative test, reconciliation.	Marketing + privacy
Partial failure / lost update	Medium / High	Correlation ID, retry state, dead-letter owner, controlled replay, reconciliation runbook.	Operations
Out-of-order status	Medium / Medium	Source timestamp, monotonic update rule, freshness display, order-of-events test.	ERP + CRM

6. Implementation and validation plan

Phase	Activities	Exit evidence
1 Discover	Inventory current interfaces, owners, data, policy, volume, and support needs.	Approved baseline; assumptions log; owners.
2 Design	Approve diagram, field map, auth, error model, objectives, privacy and retention.	Design sign-off; risk review; capacity estimate; mapping contract.
3 Build in test	Configure endpoints, secrets, transforms, queues, idempotency, logs, and alerts.	Versioned config; peer review; deployment package.
4 Validate	Run functional, negative, security, volume, recovery, and reconciliation tests.	Test results; defects/retest; acceptance against agreed targets.
5 Cut over	Pilot, compare source/target counts, go/no-go, and preserve rollback.	Change approval; support roster; rollback triggers; reconciliation.
6 Operate	Review freshness, errors, API use, credentials, mappings, incidents, and service objectives.	Runbook; incident history; service review; change log.

Minimum test cases and acceptance criteria

Scenario	Expected result / evidence
Valid Closed Won transaction	One request, one ERP order, acknowledgment linked to CRM opportunity.
Repeat the same request	No duplicate order; original result returned or logged.
Unknown ERP customer / invalid line	No order; actionable validation failure routed to an owner.
ERP timeout / recovery	Bounded retry; visible state; recovery without duplicate write.
Older status arrives late	Older timestamp cannot replace newer status; freshness visible.
Marketing opt-out during sync	Suppression follows approved target; record excluded from next audience export.
Invalid token / insufficient scope	Denied safely; no secret in logs; alert goes to access owner.
Peak volume / limit response	Approved throughput and recovery without retry storm; usage monitored.

7. Monitoring and support

- Dashboard: volume, success/failure rate, p95 duration, queue age, retry/dead-letter count, freshness by source, reconciliation variance, and Salesforce API usage. Calibrate thresholds during a pilot.
- Critical order or consent failures alert the named on-call owner. Include interface, correlation ID, entity key, error class, and safe replay steps.
- If duplication or privacy risk is suspected, pause the affected route; preserve the message; notify owners; repair and approve replay; reconcile before closure.
- Review credentials, app permissions, schema changes, retention, and recovery exercises regularly.

8. Decisions requested

Approve system-of-record ownership, consent precedence, external keys, exception ownership, and proposed service objectives. Confirm whether the ERP order exchange should be asynchronous (recommended here for resilience) or synchronous for immediate user feedback. Select middleware only after reviewing existing enterprise standards, connectors, security, support skills, and cost. Authorize discovery and a sandbox proof of concept before any implementation claim. Proposed objectives - order acknowledgement <=5 minutes p95, order status <=15 minutes p95, suppression <=1 minute where supported - are planning goals, not measured or guaranteed results.

References and sample message

[1] Salesforce Developers, OAuth and External Client Apps: https://developer.salesforce.com/docs/platform/api-rest/guide/intro-oauth-and-connected-apps.html | [2] Bulk API 2.0: https://developer.salesforce.com/docs/platform/api-asynch/guide/bulk-api-2-0.html | [3] Platform Events: https://developer.salesforce.com/docs/platform/platform-events/guide/platform-events-intro.html | [4] API Request Limits: https://developer.salesforce.com/docs/platform/salesforce-app-limits-cheatsheet/guide/salesforce-app-limits-platform-api.html | [5] Salesforce Architects, Integration Patterns: https://architect.salesforce.com/docs/architect/fundamentals/guide/integration-patterns

Official documentation accessed 31 October 2026. Validate current product behavior, limits, and org policy before implementation.

Illustrative order request envelope

This pseudo-payload shows the minimum matching and traceability fields; it is not a valid API request and contains no real customer data. Map field names and validation to the approved interface contract.

```
{
  "messageId": "SIM-ORD-00041",
  "correlationId": "SIM-CRM-OPP-0087",
  "eventType": "OrderRequested",
  "occurredAt": "2026-10-31T10:00:00Z",
  "crmOpportunityId": "SIM-OPP-0087",
  "erpCustomerId": "SIM-CUST-1042",
  "currency": "USD",
  "lines": [{"sku": "SIM-CHAIR-01", "quantity": 12}],
  "schemaVersion": "1.0"
}
```

Error recovery example

If the ERP times out, keep the same message and idempotency key in the durable queue, apply bounded retries, and alert the integration owner after the agreed threshold. If the ERP accepted the order but the acknowledgement was lost, replay must resolve to the existing order using the external reference, not create another. Record the final ERP order ID and reconcile the source opportunity against the ERP order before closing the incident. Never replay a consent or order message after a mapping change without checking its version and business effect.

Operational evidence to retain

- Interface name, message/correlation ID, source/target timestamps, mapping version, outcome, and safe error category.
- Retry count, dead-letter reason, approver for replay, reconciliation result, and incident/change reference.
- Do not log access tokens, passwords, full payment data, or unnecessary personal data.